

Pure Transition Pi-Bases Calibrate the Boundary Cases

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Source Problem

The source paper is Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257. It proves that a von Neumann algebra has BCP/UBCP if and only if it is atomic and separably represented (Theorem 3.8), and that a unital commutative C^* -algebra $C(K)$ has BCP/UBCP if and only if $K = P(C(K))$ has a countable π -basis (Corollary 3.9). It then asks:

$\{U_\alpha\}_{\alpha \in A}$ of K is called a π -basis (weak basis) for K if every nonempty open subset of K contains some U_α in $\mathcal{U}$ [37, Page 15].

In the setting of commutative C^* -algebra, [25, Theorem 2.1] can be reformulated as follows. Note that [25, Theorem 2.1] only deals with real continuous functions. The complex case is a direct consequence of the real case.

Corollary 3.9. *Let $\mathcal{A}$ be a unital commutative C^* -algebra. Then the followings are equivalent:*

- (1) $\mathcal{A}$ has the BCP;
- (2) the pure state space $\mathfrak{P}(\mathcal{A})$ has a countable π -basis;
- (3) $\mathcal{A}$ has the UBCP.

In particular, there exist commutative C^* -algebras containing no minimal projections but having the BCP. For example, the uniform norm closure of the linear span of Haar functions [24, p.150].

We conclude this section by the following question.

Question 3. *It will be interesting to find a non-commutative version of Corollary 3.9, that is, how to characterize those C^* -algebras $\mathcal{A}$ having BCP/UBCP.*

Transition Functionals

For a C^* -algebra A , define

$$T(A) = \text{ext}(B_{A^*}),$$

the extreme points of the dual unit ball, with the relative weak-star topology inherited from A^* . These are the norm-one pure functionals. In an irreducible representation they are the noncommutative analogue of rank-one coefficient functionals

$$a \longmapsto \langle \pi(a)\xi, \eta \rangle.$$

The point of this packet is to check that $T(A)$, rather than only the positive pure state space, matches the two boundary cases already solved in the source paper.

Statement

Theorem 1 (commutative calibration). *Let K be a compact Hausdorff space. Then $T(C(K))$ has a countable π -base if and only if K has a countable π -base. Consequently, for a unital commutative C^* -algebra $A = C(K)$, the following are equivalent:*

$$A \text{ has BCP,} \quad T(A) \text{ has a countable } \pi\text{-base,} \quad A \text{ has UBCP.}$$

Theorem 2 (von Neumann transition obstruction). *Let M be a von Neumann algebra. If $T(M)$ has a countable π -base, then M is atomic and separably represented. Consequently M has BCP and UBCP by the source paper's Theorem 3.8.*

Corollary 1. *Every von Neumann algebra failing BCP in the source theorem has transition functional space $T(M)$ with no countable π -base.*

Idea of the Proof

For $C(K)$, the Riesz representation theorem identifies the extreme points of the dual unit ball with the point masses and phases:

$$T(C(K)) \cong \mathbb{T} \times K, \quad (\lambda, x) \mapsto \lambda \delta_x.$$

Product with the metrizable circle preserves the existence of a countable π -base.

For von Neumann algebras, the proof is a transition-functional version of the pure-state obstruction. In an atomless von Neumann algebra, the absolute value of every pure functional is a singular pure state. Given countably many proposed transition-open sets, choose one pure functional from each. A common projection kills all of their absolute values. The transition-open set of functionals that see that projection is nonempty, but none of the chosen open sets is contained in it.

In the atomic but nonseparably represented case, uncountably many central summands or uncountably many orthogonal rank-one projections give uncountably many disjoint nonempty transition-open sets, ruling out a countable π -base.

The Commutative Case

Proof of Theorem 1. By the Riesz representation theorem, $C(K)^*$ is the regular Borel measure space $M(K)$. The extreme points of the unit ball of $M(K)$ are exactly

$$\lambda \delta_x, \quad \lambda \in \mathbb{T}, x \in K.$$

The map $\mathbb{T} \times K \rightarrow T(C(K))$, $(\lambda, x) \mapsto \lambda \delta_x$, is a homeomorphism for the relative weak-star topology.

If K has a countable π -base (U_n) , and (V_m) is a countable base for $\mathbb{T}$, then the sets

$$V_m \times U_n$$

form a countable π -base for $\mathbb{T} \times K$.

Conversely, suppose $\mathbb{T} \times K$ has a countable π -base (W_n) . For every nonempty open $U \subset K$, the set $\mathbb{T} \times U$ is a nonempty open subset of $\mathbb{T} \times K$. Choose n with

$$W_n \subset \mathbb{T} \times U.$$

The projection of W_n onto K is a nonempty open subset of K contained in U . Therefore the countable family of nonempty projections of the W_n 's is a π -base for K .

The final equivalence follows from the source paper's Corollary 3.9. $\square$

A Common Projection Lemma

Lemma 1 (common projection for singular states). *Let N be a nonzero von Neumann algebra and let (φ_n) be countably many singular states on N . Then there is a nonzero projection $p \in N$ such that*

$$\varphi_n(p) = 0 \quad (n \geq 1).$$

Proof. Fix a normal state ω on N . For a singular positive functional φ , there are projections $e_i \uparrow 1$ with $\varphi(e_i) = 0$. Since ω is normal, for every $\varepsilon > 0$ there is a projection e with

$$\varphi(e) = 0, \quad \omega(1 - e) < \varepsilon.$$

Apply this to φ_n with $\varepsilon_n = 2^{-n-2}$, obtaining projections e_n with

$$\varphi_n(e_n) = 0, \quad \omega(1 - e_n) < 2^{-n-2}.$$

Set $p = \bigwedge_n e_n$. Then

$$\omega(1 - p) \leq \sum_n \omega(1 - e_n) < \frac{1}{2},$$

so $p \neq 0$. Since $p \leq e_n$, positivity gives $\varphi_n(p) = 0$ for every n . □

Lemma 2. *If N is a nonzero atomless von Neumann algebra, then $T(N)$ has no countable π -base.*

Proof. Let (U_n) be any countable family of nonempty open subsets of $T(N)$. Choose $f_n \in U_n$. For each n , let $|f_n|$ be the absolute value of f_n in the polar decomposition of the functional. Since f_n is an extreme point of the dual unit ball, $|f_n|$ is a pure state. An atomless von Neumann algebra has no normal pure states, so $|f_n|$ is singular.

By Lemma 1, choose a nonzero projection $p \in N$ such that

$$|f_n|(p) = 0 \quad (n \geq 1).$$

The Cauchy–Schwarz inequality in the polar decomposition gives

$$|f_n(p)|^2 \leq |f_n|(p) = 0,$$

so $f_n(p) = 0$ for every n .

Choose a pure state ψ_0 on the nonzero unital C^* -algebra pNp . Its canonical extension

$$\psi(x) = \psi_0(pxp) \quad (x \in N)$$

is a pure state of N , and $\psi(p) = 1$. Hence

$$W = \{f \in T(N) : |f(p)| > 1/2\}$$

is a nonempty open subset of $T(N)$.

But $f_n \in U_n \setminus W$ for every n . Thus no U_n is contained in W , and the countable family (U_n) cannot be a π -base. □

Atomic Nonseparable Von Neumann Algebras

Lemma 3. *If a topological space contains uncountably many pairwise disjoint nonempty open subsets, then it has no countable π -base.*

Proof. If (V_n) were a countable π -base and $(O_i)_{i \in I}$ were pairwise disjoint nonempty open sets, each O_i would contain some $V_{n(i)}$. Distinct i 's require distinct $n(i)$'s, because the O_i 's are disjoint and $V_{n(i)} \neq \emptyset$. Hence I would be countable. $\square$

Lemma 4. *Let M be an atomic von Neumann algebra which is not separably represented. Then $T(M)$ has no countable π -base.*

Proof. Write

$$M \cong \prod_{i \in I} \mathcal{B}(H_i).$$

If I is uncountable, let z_i be the central projection onto the i -th summand. The sets

$$O_i = \{f \in T(M) : |f(z_i)| > 1/2\}$$

are nonempty and open. They are pairwise disjoint: if $i \neq j$, the restriction of f to the abelian algebra generated by z_i, z_j has norm at most one, so $|f(z_i)| + |f(z_j)| \leq 1$. Lemma 3 applies.

Now suppose I is countable but M is not separably represented. Then some summand is $\mathcal{B}(H)$ with H nonseparable. Choose an uncountable orthonormal family $(\xi_\alpha)_{\alpha \in \Gamma}$ in H , and let p_α be the rank-one projection onto $\mathbb{C}\xi_\alpha$. The sets

$$W_\alpha = \{f \in T(\mathcal{B}(H)) : |f(p_\alpha)| > 1/2\}$$

are nonempty open subsets of $T(\mathcal{B}(H))$. They are pairwise disjoint, again because the restriction of a norm-one functional to the abelian algebra generated by two orthogonal rank-one projections has norm at most one.

The transition functional space $T(\mathcal{B}(H))$ sits as a nonempty open subspace of $T(M)$, determined by the corresponding central projection. Thus $T(M)$ also contains uncountably many pairwise disjoint nonempty open sets. Lemma 3 applies. $\square$

Proof of the Von Neumann Calibration

Proof of Theorem 2. We prove the contrapositive. Suppose M is not atomic and separably represented.

If M has a nonzero atomless central summand $N = zM$, then $T(N)$ is identified with the open subspace

$$\{f \in T(M) : |f(z)| > 1/2\}.$$

Indeed, an extreme norm-one functional is supported on exactly one side of the central decomposition $M = zM \oplus (1 - z)M$. By Lemma 2, this open subspace has no countable π -base, so $T(M)$ has no countable π -base.

If M is atomic but not separably represented, then Lemma 4 gives the same conclusion.

Therefore $T(M)$ having a countable π -base forces M to be atomic and separably represented. The source paper's Theorem 3.8 then gives BCP and UBCP. $\square$

Proof of Corollary 1. The source theorem says that a von Neumann algebra fails BCP exactly when it is not atomic and separably represented. Apply Theorem 2. $\square$

Relation to the Full C*-Problem

This packet is not a full solution of Question 3. It says that the transition functional space is a viable noncommutative replacement for the commutative pure-state space at the two known boundaries:

$$C(K) : \quad T(C(K)) \cong \mathbb{T} \times K,$$

$$\text{von Neumann } M : \quad T(M) \text{ countable } \pi\text{-base} \Rightarrow M \text{ in the positive source class.}$$

The missing arbitrary C^* -step is still a local bump theorem:

$$T(A) \text{ has enough countable local coefficient patches} \implies A \text{ has UBCP.}$$

The split-interval counterexample from the lane shows that such patches cannot require separable principal right ideals. The positive mechanism must allow scalar or rank-one coefficient bumps over locally non-second-countable sets.

Novelty and Search Bounds

Local indexes were searched for arXiv:2311.14257, Corollary 3.9, Theorem 3.8, pure state space, pure transition functionals, extreme dual ball, π -base, atomless von Neumann algebra, singular pure state, and the previous lane-19 C^* -BCP packets. The source paper contains the von Neumann BCP/UBCP classification and the commutative pure-state theorem, but no local packet or source theorem recorded the transition-functional boundary calibration in the form proved here.

Human Review Focus

Reviewers should check:

1. the homeomorphism $T(C(K)) \cong \mathbb{T} \times K$;
2. the assertion that the absolute value of a pure functional is a pure state;
3. the use of singularity for pure states on atomless von Neumann algebras;
4. the Cauchy-Schwarz step $|f(p)|^2 \leq |f|(p)$;
5. the uncountable disjoint-open-set arguments for atomic nonseparable von Neumann algebras.

References

- [1] Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257, 2023.
- [2] J. Dixmier, *C^* -Algebras*, North-Holland, 1977.
- [3] S. Sakai, *C^* -Algebras and W^* -Algebras*, Springer, 1971.
- [4] Bruce Blackadar, *Operator Algebras: Theory of C^* -Algebras and Von Neumann Algebras*, Encyclopaedia of Mathematical Sciences 122, Springer, 2006.

- [5] Run `fa_banach_001`, lane 19, *Pure-State Pi-Bases Obstruct the Negative Von Neumann Cases*, solution packet `2311.14257_von_neumann_pure_state_pi_base_obstruction`, 2026.
- [6] Run `fa_banach_001`, lane 19, *Pure-State Pi-Base Does Not Force a Separable Right-Support Net*, solution packet `2311.14257_pure_pi_base_not_support_net_split_interval`, 2026.